Qu. 1. a) Find the Cartesian and cylindrical polar coordinates of the point P which has spherical coordinates $(3, \pi/4, \pi/3)$.

b) Find the spherical polar coordinates of the point Q which has Cartesian coordinates $(1, 1, 1)$.

c) The surface, S, of a sphere of radius R, centre the origin, can be expressed in spherical polars coordinates as the set $\{(\rho, \varphi, \theta) : \rho = R, 0 \leqslant \varphi \leqslant \pi, 0 \leqslant \theta < 2\pi\}$. Find expressions for S in terms of Cartesian and cylindrical polar coordinates.

d) The surface, C, of a cylinder of radius R with axis lying along the z-axis can be expressed in terms of cylindrical polar coordinates as the set $\{(r, \theta, z) : r = R, 0 \leqslant \theta < 2\pi, -\infty < z < \infty\}$. Find expressions for C in terms of Cartesian and spherical polar coordinates.

e) Let K be the surface of an infinite cone with circular cross section, vertex at the origin and axis lying along the positive z-axis. If the angle between the z-axis and the surface of the cone is α, find expressions for K in terms of Cartesian, spherical and cylindrical polar coordinates.

Qu. 2. Verify that:

a) the element of area $dA = dx\,dy$ is equal to $r\,dr\,d\theta$ in polar co-ordinates;

b) the element of volume $dV = dx\,dy\,dz$ is equal to $r\,dr\,d\theta\,dz$ in cylindrical polar co-ordinates;

c) the element of volume $dV = dx\,dy\,dz$ is equal to $\rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta$ in spherical polar co-ordinates.

Qu. 3. Evaluate the following double integral as a repeated integral: $\displaystyle\iint_R (x^2 + y^2)\,dA$ where R is the sector $\{(r, \theta) : 1 < r < 2, \pi/3 < \theta < \pi/2\}$ [Hint: use polar coordinates]

Qu. 4. a) The domain bounded by the surface of a paraboloid $z = 2 - x^2 - y^2$ and that of a cone $z^2 = x^2 + y^2$ is given by $D = \{x, y, z : \ x^2 + y^2 \leqslant 1, \sqrt{x^2 + y^2} \leqslant z \leqslant 2 - x^2 - y^2\}$. Find its volume using the appropriate coordinate system.

Qu. 5. Evaluate each of the following triple integrals as a repeated integral using an appropriate coordinate systems:

a) $\displaystyle\iiint_R z e^{-(x^2+y^2+z^2)}\,dV$, where $R = \{(x, y, z) : -\infty < x, y < \infty, 0 \leqslant z < 1\}$.

b) $\displaystyle\iiint_S x^2\,dV$, where S is the unit ball $\{(\rho, \varphi, \theta) : \rho < 1\}$.

Qu. 6. Find the volume between the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 1$ that lies in the first octant (i.e. $x > 0$, $y > 0$, $z > 0$) using cylindrical coordinates.